

Envy-Free Traffic Control for City-Wide Emergency Routing via Hierarchical Multi-Agent Actor–Critic

First Author^{1,2*}, Second Author^{2,3†} and Third Author^{1,2†}

^{1*}Department, Organization, Street, City, 100190, State, Country.

²Department, Organization, Street, City, 10587, State, Country.

³Department, Organization, Street, City, 610101, State, Country.

*Corresponding author(s). E-mail(s): iauthor@gmail.com;

Contributing authors: iauthor@gmail.com; iiiauthor@gmail.com;

[†]These authors contributed equally to this work.

Abstract

Efficient and equitable traffic signal control is critical for large-scale urban networks, particularly under emergency scenarios where routing priority must be dynamically enforced without degrading overall system performance. Existing approaches either lack global coordination at scale, assume fixed intersection topologies incompatible with heterogeneous city road networks, or omit explicit mechanisms for emergency vehicle prioritization and fairness. We propose a hierarchical actor–critic framework that addresses these gaps through four integrated contributions: topology-aware network partitioning via the Louvain community detection algorithm, a dynamically adaptive intersection model supporting variable neighbor counts and phase sets, multi-EMV routing with proactive corridor pre-clearance, and an envy-free regularization mechanism that ensures equitable signal allocation across intersections. The framework is validated on synthetic benchmarks and, for the first time, on the real urban road network of Coimbatore, Tamil Nadu, under stochastic accident injection and multiple simultaneous EMV dispatch scenarios. Experimental results demonstrate improvements in EMV throughput, post-preemption recovery, and overall traffic efficiency over state-of-the-art hierarchical and multi-agent baselines.

Keywords: Traffic Signal Control, Multi-Agent Reinforcement Learning, Emergency Vehicle Routing, Community Detection, Louvain Algorithm, Fairness-Aware Optimization, Hierarchical Reinforcement Learning

1 Introduction

Urban traffic congestion represents one of the most pressing infrastructure challenges of the modern era, with cascading consequences for emergency response efficiency, economic productivity, and public health. Among the various components of urban traffic management, the coordinated control of traffic signals at intersections and the rapid, unimpeded passage of emergency vehicles (EMVs)—including ambulances, fire trucks, and police vehicles—stand out as problems of acute societal importance. The urgency is quantified starkly by medical literature: the survivor rate from sudden cardiac arrest without treatment drops 7–10% for every minute elapsed, and survival probability approaches zero beyond eight minutes [2]. Yet despite decades of research, existing frameworks for traffic signal control (TSC) and emergency vehicle management remain siloed, individually limited in scale, and largely blind to fairness considerations and real-world disruption scenarios.

Strengths and Limitations of EMVLight

EMVLight [2] is a landmark contribution to the joint problem of EMV routing and traffic signal pre-emption. Its core innovation is the formulation of each intersection as a reinforcement learning (RL) agent under a multi-agent advantage actor-critic (MA2C) framework, with agents classified dynamically into three types—primary pre-emption, secondary pre-emption, and normal—depending on their proximity to the dispatched EMV. A spatial discount factor α in the adjusted reward function ensures that each agent pays proportionally less attention to the rewards of more distant agents, enabling decentralized execution without sacrificing coordination awareness. A decentralized dynamic Dijkstra routing heuristic allows each intersection to update its estimated time of arrival (ETA) to the EMV destination using only locally observed intra-link travel times, reducing communication overhead compared to centralized re-routing. On synthetic and real-world benchmarks including a Manhattan 16×3 network and Hangzhou 4×4, EMVLight achieves up to a 42.6% reduction in EMV travel time and a 23.5% reduction in average non-EMV travel time compared to existing methods.

Despite these strengths, EMVLight has five important limitations. **(i)** The reward structure and agent-type classification assume exactly *one* EMV in transit at any given time; the pre-emption hierarchy becomes undefined when multiple EMVs are simultaneously active. **(ii)** The routing mechanism is fundamentally *reactive*: ETA updates depend on intra-link travel times observed at the moment of update, so the framework cannot anticipate downstream congestion before it reaches the EMV’s immediate neighborhood. **(iii)** Coordination is limited to immediate neighbors; no higher-level regional abstraction exists, making network-wide traffic patterns invisible to individual agents. **(iv)** The pressure-based reward optimizes aggregate network efficiency but provides no fairness guarantees: intersections that are structurally disadvantaged by their position in the network may receive systematically inferior service with no corrective mechanism. **(v)** The framework assumes a fixed network topology and does not model accident-induced capacity reduction—precisely the scenario most likely to trigger an EMV dispatch while simultaneously degrading the route the EMV must traverse.

Strengths and Limitations of HiLight

HiLight [1] addresses the scalability challenge that EMVLight does not confront. Its hierarchical RL (HRL) structure separates decision-making into a global *Meta-Policy* and a decentralized *Sub-Policy*. The Meta-Policy employs a Transformer encoder to exchange information across subregions and an LSTM to model temporal dependencies in regional traffic states, generating a global sub-goal vector broadcast to all Sub-Policy agents. An adversarial training mechanism—wherein the Meta-Policy sets increasingly challenging sub-goals while the Sub-Policy is trained to surpass them—provides principled joint optimization and avoids the reward misalignment common in naive hierarchical approaches. Evaluated across five standard benchmarks and three large-scale Manhattan configurations (Peak Transition, Adverse Weather, and Holiday Rush flows), HiLight outperforms eleven baselines and demonstrates performance improvements that scale positively with network size, reaching 2,668 intersections in its largest scenario.

However, HiLight has five complementary limitations. **(i)** Network partitioning is performed using a *fixed, manual grid*: for Manhattan2668 a 4×15 grid is imposed, producing 60 regions irrespective of the underlying road topology. Grid boundaries may cut through densely connected subgraphs and group intersections that share little traffic exchange, introducing noise into the Meta-Policy’s regional state representations. **(ii)** The Sub-Policy’s Graph Attention Concat (GAC) module assumes *exactly four neighbors* and *exactly eight signal phases* per intersection. Real urban networks contain three-way junctions, five-arm roundabouts, and rotary intersections, each requiring a different phase set; HiLight’s fixed structural assumptions preclude direct deployment on such heterogeneous road maps without manual preprocessing. **(iii)** HiLight contains no EMV prioritization mechanism: there is no agent-type classification, pre-emption logic, or routing integration. **(iv)** Like EMVLight, HiLight offers no intersection-level fairness mechanism and no treatment of supply-side disruptions such as accidents. **(v)** HiLight’s evaluation is confined to simulated environments with demand-side variability; its deployability on heterogeneous real city road networks with stochastic incident injection has not been demonstrated.

The Present Work

We propose a framework that resolves the above gaps through five integrated contributions.

(1) Topology-Aware Community Detection via Louvain Algorithm. We replace HiLight’s manual grid partitioning with the *Louvain algorithm* [3], which partitions the road network graph into communities by iteratively maximizing the modularity Q :

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \quad (1)$$

where A_{ij} is the weighted adjacency matrix of the road network (edge weights reflect traffic volume or travel time), k_i and k_j are the weighted degrees of nodes i and j , m is the sum of all edge weights, and $\delta(c_i, c_j)$ equals 1 if i and j belong to the

same community and 0 otherwise. The result is a set of communities corresponding to densely internally connected subgraphs—natural traffic catchment zones where vehicles predominantly circulate internally and cross boundaries only for longer trips. Community boundaries tend to align with major arterials and natural barriers, making them meaningful coordination interfaces for both general TSC and EMV corridor pre-clearance.

We compare Louvain against the *Leiden algorithm* [4], which guarantees internally connected communities by adding a refinement phase after each modularity-improvement step. In our empirical evaluation on the road networks of Manhattan and Coimbatore, Louvain consistently achieves higher modularity scores and more stable community assignments across random initializations. Road networks are naturally sparse and planar, which causes Leiden to produce smaller, more fragmented communities that reduce the effective coordination range of the Meta-Policy. We therefore adopt Louvain as our default partitioning method; a comparative analysis of modularity, normalized mutual information (NMI), and downstream TSC performance is presented in Section ??.

(2) Dynamic Intersection Topology and Adaptive Phase Assignment.

Each intersection i is characterized by its actual number of incoming and outgoing links $\{l_i^{\text{in}}, l_i^{\text{out}}\}$ as determined by the road network map. The permissible signal phase set $\mathcal{A}_i$ is generated automatically from all non-conflicting traffic movement combinations for that intersection’s specific geometry. The Graph Attention module in the Sub-Policy handles variable-degree neighborhoods: attention scores are computed over $|\mathcal{N}(i)|$ actual neighbors (which varies by intersection), and zero-padding is applied only at the feature aggregation stage rather than being embedded in the network architecture. This design enables direct deployment on heterogeneous networks—including the Coimbatore city road map, which contains three-way junctions, four-way intersections, rotary intersections, and flyover merge points—without manual preprocessing or network simplification.

(3) Hierarchical Multi-EMV Routing with Proactive Corridor Pre-Clearance. We extend EMVLight’s multi-class agent design into the hierarchical framework and support multiple simultaneously active EMVs. Each active EMV induces a dynamic agent-type classification over the intersections in its downstream corridor. When two EMV corridors overlap, a priority resolution mechanism arbitrates pre-emption sub-goals based on EMV type and estimated remaining travel time. The Meta-Policy is augmented with an EMV awareness module that identifies affected communities along each active EMV’s projected route and issues corridor pre-clearance sub-goals in anticipation of the EMV’s arrival, elevating routing from a reactive link-by-link heuristic to proactive community-level planning.

(4) Envy-Free Intersection-Level Fairness. We introduce an envy-free regularization term $\mathcal{L}_{\text{envy}}$ into the Sub-Policy’s Actor-Critic loss. For neighboring agent pairs within the same Louvain community, the term penalizes configurations in which one agent would systematically prefer its neighbor’s resource allocation over its own. The total Sub-Policy loss becomes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{AC}} + \lambda_e \mathcal{L}_{\text{envy}}, \quad (2)$$

where $\mathcal{L}_{AC}$ is the standard Actor-Critic loss and λ_e balances efficiency against equity. This mechanism is particularly valuable during EMV pre-emption, when primary and secondary pre-emption agents receive systematically different reward signals from normal agents. The envy-free penalty accelerates post-preemption service restoration, addressing a residual congestion effect that neither EMVLight nor HiLight manages explicitly.

(5) Accident-Driven Traffic Control on the Coimbatore City Map.

We validate the complete framework on a real urban road network extracted from Coimbatore, Tamil Nadu, under stochastic accident injection scenarios with multiple simultaneous EMV dispatches. To our knowledge, this is the first evaluation of a hierarchical RL-based TSC framework with integrated EMV routing on an Indian city road network, and the first to treat the joint problem of accident-induced congestion and EMV passage in a unified learning framework.

The remainder of this paper is organized as follows. Section 2 reviews relevant literature. Section ?? presents the problem formulation and community detection methodology. Section ?? describes the proposed framework architecture. Section ?? details experimental setup and results. Section 3 concludes the paper.

2 Literature Survey

2.1 Conventional EMV Routing and Traffic Signal Pre-Emption

Early approaches to EMV routing treat the road network as a static graph and apply Dijkstra’s shortest-path algorithm or its variants—A* search with Manhattan distance heuristics—to compute an optimal route prior to dispatch [2]. Once a route is fixed, traffic signal pre-emption is deployed independently: green phases along the planned route are extended to allow unimpeded EMV passage, a strategy known as Green Wave control. While straightforward, this decoupled approach is fundamentally limited. Route optimality degrades as traffic conditions evolve during the EMV’s journey, and pre-emption strategies calibrated to a static route cannot adapt when the optimal path changes. The negative secondary effects on non-EMVs—sudden stops, lane blockages, and delayed signal restoration—are also not addressed. Dynamic routing extensions, such as repeatedly re-running A* at fixed intervals, reduce route staleness but incur quadratic communication costs as network size grows, and still do not solve the coupling between routing decisions and their effect on signal timing.

2.2 Reinforcement Learning for Traffic Signal Control

The application of deep reinforcement learning to TSC has progressed from isolated single-intersection controllers using Q-learning to large-scale multi-agent systems capable of coordinating hundreds of intersections simultaneously. Q-learning-based methods such as PressLight [2] incorporate max-pressure control into reward design, selecting the signal phase with the highest pressure to smooth congestion. CoLight extends this by using graph attentional networks to facilitate communication between neighboring intersections. The actor-critic family of methods avoids the curse of

dimensionality that limits Q-learning in multi-agent settings: under the MA2C framework of [2], each intersection maintains a local policy (actor) and value function (critic), coordinated via spatially discounted shared rewards. Despite strong performance on benchmark networks, none of these methods were designed with EMV pre-emption in mind, and none provide fairness guarantees at the intersection level.

2.3 EMVLight: Joint Routing and Pre-Emption via MA2C

EMVLight [2] is the first framework to jointly address EMV dynamic routing and network-wide traffic signal control in a single reinforcement learning formulation. By classifying each intersection agent into one of three types depending on its proximity to the dispatched EMV, and designing distinct reward functions for each type, EMVLight learns cooperative signal phasing strategies that reduce both EMV travel time and the average travel time of non-EMVs. The decentralized dynamic Dijkstra heuristic enables efficient route updates at each intersection using only local information. A pressure-based intersection reward function, adapted from PressLight, provides a robust training signal. The framework achieves up to a 42.6% reduction in EMV travel time on real-world and synthetic benchmarks. Its primary structural constraints—single EMV assumption, reactive routing, no regional abstraction, no fairness mechanism—are identified as research gaps motivating the present work.

2.4 HiLight: Hierarchical RL for Large-Scale TSC

HiLight [1] demonstrates that the scalability limitations of flat multi-agent approaches can be overcome by structuring the policy hierarchy into a global Meta-Policy and decentralized Sub-Policy agents. The Meta-Policy uses a Transformer encoder to aggregate information across manually defined subregions and an LSTM to capture temporal traffic dynamics, generating sub-goal vectors that guide Sub-Policy agents toward globally consistent objectives. An adversarial training mechanism wherein the Meta-Policy sets progressively challenging sub-goals while the Sub-Policy is trained to surpass them provides stable joint optimization and avoids reward misalignment. HiLight reaches state-of-the-art performance on networks up to 2,668 intersections, with performance advantages that grow with network size. Its constraints of fixed grid partitioning, fixed intersection topology, and absence of EMV or fairness mechanisms are the direct starting points for our proposed extensions.

2.5 Community Detection in Transportation Networks

Graph community detection methods identify densely connected subgraphs within a network, and have been applied to transportation problems including traffic zone delineation, origin–destination demand estimation, and transit network design. The Louvain algorithm [3] optimizes modularity Q (Equation 1) through a two-phase iterative procedure: in the first phase, each node is tentatively moved to the neighboring community that produces the greatest increase in Q ; in the second phase, a new graph is constructed by collapsing each community into a single node, and the procedure repeats. The algorithm terminates when no move improves Q . The Leiden algorithm [4] refines Louvain by adding a node refinement step that guarantees all

output communities are internally connected, correcting a theoretical weakness in Louvain where nodes can become disconnected from their own community. In practice, Leiden communities on road networks tend to be smaller and more fragmented than Louvain communities due to the sparse, planar nature of road graphs. Our empirical comparison—using modularity, NMI between partitions, and downstream TSC performance as metrics—confirms Louvain as the preferred method for the traffic network partitioning task.

2.6 Fairness in Multi-Agent Traffic Control

The notion of envy-freeness originates in fair division theory: an allocation is envy-free if no agent prefers any other agent’s allocation to its own. Recent work in multi-agent reinforcement learning has begun to incorporate fairness objectives, noting that aggregate reward maximization can produce policies that systematically disadvantage agents in structurally unfavorable network positions. In the TSC context, this manifests as chronic under-service at intersections downstream of persistent bottlenecks. Fairness-aware TSC remains an open problem: pressure-based reward designs optimize flow efficiency but carry no equity guarantees, and none of the major TSC frameworks—including EMVLight and HiLight—incorporate an explicit fairness mechanism. Introducing envy-free regularization into the actor-critic training objective is, to our knowledge, a novel contribution to this literature.

3 Future Works

3.1 Enhanced Emergency Vehicle Routing

The routing mechanism presented in this work improves on EMVLight’s reactive dynamic Dijkstra heuristic by integrating route planning into the hierarchical Meta-Policy. However, several directions remain open. At each Meta-Policy decision step, a candidate route set is generated from current regional pressure readings and the known destination, and sub-goals are issued to Sub-Policy agents along candidate corridors to reduce queue lengths before EMV arrival. This transforms routing from a two-horizon problem (current link and next link) into a multi-horizon planning problem aligned with the hierarchical structure. Future work will extend this to settings where the destination is unknown at dispatch time—for example, in mass-casualty events where the nearest available hospital is itself dynamically determined—and to scenarios with more than two simultaneously active EMVs whose corridors intersect repeatedly. The Louvain community structure provides a natural abstraction for corridor planning at the boundary level: because inter-community transitions represent sparse bottleneck links, the Meta-Policy can reason about sequences of community transitions rather than individual links, substantially reducing the planning horizon while preserving global routing awareness.

3.2 Envy-Free Traffic Control

The envy-free regularization introduced in this work operates within individual Louvain communities during training. Future work will investigate adaptive scheduling

of λ_e across training: starting with a high efficiency weight and gradually increasing the fairness weight as the policy matures may accelerate convergence while preserving equity in the final policy. Additionally, the definition of envy used here is pairwise and local; extending it to group envy—where a set of intersections collectively prefers another group’s allocation—would capture more subtle structural inequities in large networks. The interaction between the envy-free penalty and the EMV pre-emption reward also warrants further study: during active EMV passage, pre-emption agents and normal agents necessarily receive asymmetric rewards, and calibrating λ_e dynamically based on EMV proximity could allow the system to temporarily suspend fairness constraints along the active corridor while preserving them network-wide.

3.3 Accident-Based Traffic Control on Real City Maps

The accident injection model used in this work assigns a fixed severity parameter κ to each incident and samples accident locations according to a crash probability distribution derived from historical report data. Future work will model accidents with evolving severity: an initial partial blockage ($\kappa < 1$) that transitions to complete blockage ($\kappa = 1$) as secondary vehicles arrive and then gradually clears as the incident is resolved. This temporal evolution of supply disruption is a qualitatively harder problem for the Meta-Policy, requiring it to track incident lifecycle stages rather than treating each accident as a static constraint. Evaluation on the Coimbatore city map will be extended to multi-district scenarios in which accidents in one part of the city cascade congestion into adjacent zones, stressing the inter-community coordination capacity of the Meta-Policy. The combination of accident-induced rerouting, multiple active EMVs, and envy-free fairness enforcement represents the most realistic and demanding test scenario for the complete framework, and constitutes the primary experimental target for the next phase of this research.

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